

DAA ASSIGNMENT

OUTPUT SCREENSHOTS:

PROGRAM CODE OUTPUT:

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LOGISTICS RESOURCE ALLOCATION SYSTEM
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Number of Packages : 10

Enter Vehicle Weight Capacity (kg): 70
Enter Vehicle Volume Capacity (L): 500

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PACKAGE DATA
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ID      Weight  Volume  Value  Priority
P1       20      10    Rs.100  Medium
P2       40      20    Rs.200  High
P3       30      15    Rs.150  Low
P4       50      25    Rs.300  High
P5       10       5     Rs.80   Medium
P6       60      30    Rs.250  Low
P7       25      12    Rs.180  High
P8       35      18    Rs.220  Medium
P9       15       7     Rs.120  High
P10      45      22    Rs.270  Low

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DYNAMIC PROGRAMMING RESULT
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Selected Packages:
P2 Weight=40 kg Volume=20 L Value=Rs.200 Priority=High
P1 Weight=20 kg Volume=10 L Value=Rs.100 Priority=Medium
P10 Weight=45 kg Volume=22 L Value=Rs.270 Priority=Low

Total Packages : 3
Total Weight : 105 kg
Total Volume : 52 L
Total Value : Rs.570
Weight Utilization : 150.00%
Volume Utilization : 10.40%

Execution Time: 0.0000 ms
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=====
GREEDY RESULT
=====

Selected Packages:
P9 Weight=15 kg Volume=7 L Value=Rs.120 Priority=High
P2 Weight=40 kg Volume=20 L Value=Rs.200 Priority=High
P1 Weight=20 kg Volume=10 L Value=Rs.100 Priority=Medium
P3 Weight=30 kg Volume=15 L Value=Rs.150 Priority=Low

Total Packages : 4
Total Weight : 105 kg
Total Volume : 52 L
Total Value : Rs.570
Weight Utilization : 150.00%
Volume Utilization : 10.40%

Execution Time: 0.0000 ms

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HYBRID RESULT
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Selected Packages:
P9 Weight=15 kg Volume=7 L Value=Rs.120 Priority=High
P7 Weight=25 kg Volume=12 L Value=Rs.180 Priority=High
P1 Weight=20 kg Volume=10 L Value=Rs.100 Priority=Medium
P10 Weight=45 kg Volume=22 L Value=Rs.270 Priority=Low

Total Packages : 4
Total Weight : 105 kg
Total Volume : 51 L
Total Value : Rs.670
Weight Utilization : 150.00%
Volume Utilization : 10.20%

Execution Time: 0.0000 ms
```

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                        ALGORITHM COMPARISON
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Algorithm           Execution Time
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Dynamic Programming  0.0000 ms
Greedy               0.0000 ms
Hybrid              0.0000 ms

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                        CONCLUSION
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Dynamic Programming provides an optimal solution
for the weight-based knapsack formulation.

Greedy provides a fast approximate solution using
value and priority based package selection.

Hybrid combines greedy selection with local
improvement to obtain a better practical solution.

The Hybrid approach is suitable for large-scale
logistics resource allocation because it balances
solution quality and computational efficiency.

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Process exited after 18.45 seconds with return value 0
Press any key to continue . . . |
```

IMPLEMENTATION OUTPUT :

 **LogiOpt**

NAVIGATION

Dashboard

Packages

Vehicles

Optimization

ANALYSIS

Performance

Reports

Optimization Engine

DP • Greedy • Hybrid

Logistics Resource Allocation

Intelligent package selection and vehicle optimization

 Total Packages
11

 Best Delivery Value
₹1550

 Weight Used
243 kg

 Capacity Utilization
97.2%

Vehicle Configuration

Define transportation constraints

Vehicle Weight Capacity

500 kg

Operational Condition

Capacity Constrained

Package Management

Add packages to the optimization dataset

Package Weight (kg)

Example: 25

Delivery Value (₹)

Example: 500

System Ready

LogiOpt

NAVIGATION

Dashboard

Packages

Vehicles

Optimization

ANALYSIS

Performance

Reports

Optimization Engine

DP • Greedy • Hybrid

Package Management

Add packages to the optimization dataset

Package Weight (kg)

Example: 25

Delivery Value (₹)

Example: 500

Priority

Medium Priority

+ Add Package

Load Sample Dataset

Clear Dataset

Package added successfully.

Optimization Algorithms

Select an algorithm to allocate packages

Dynamic Programming

Optimal solution using 0/1 Knapsack

Greedy Algorithm

Fast selection using value efficiency

Hybrid Algorithm

Greedy selection with local improvement

Compare All Algorithms

Optimization Result

Hybrid optimization completed

Algorithm

Hybrid

Packages Selected

7

Total Weight

245 kg

Total Value

₹2000

Execution Time

0.1000 ms

Vehicle Capacity Utilization

98.0%

Selected Packages

Packages chosen for transportation

PACKAGE	WEIGHT	VALUE	PRIORITY
P9	15 kg	₹120	High
P12	90 kg	₹1000	Medium
P7	25 kg	₹180	High
P4	50 kg	₹300	High
P5	10 kg	₹80	Medium
P8	35 ka	₹220	Medium

Algorithm Performance Comparison

Compare solution quality and computational performance

Dynamic Programming

Value ₹2030

Utilization 99.2%

Greedy

Value ₹1980

Utilization 100.0%

Hybrid

Value ₹2000

Utilization 98.0%

ALGORITHM	PACKAGES	WEIGHT	TOTAL VALUE	UTILIZATION	TIME
Dynamic Programming	6	248 kg	₹2030	99.20%	0.1000 ms
Greedy	7	250 kg	₹1980	100.00%	0.0000 ms
Hybrid	7	245 kg	₹2000	98.00%	0.1000 ms

Logistics Resource Allocation System • Dynamic Programming • Greedy • Hybrid